

Paper OOL-9: Protocell Integration, Replication, and Error Thresholds: The First Heritable Container Gate

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Abstract

This paper is the integration gate. A protocell claim requires bounded copying error, retained internal chemistry, coupled energy throughput, and a boundary whose persistence covaries with its internal state. The paper combines Eigen–Orgel error-threshold logic with a conservative protocell diagnostic and explicitly reports numerical fragility where it appears. That restraint is intentional: Paper 9 defines what must be true before Paper 10 can introduce parasitic exploitation.

1 Introduction

Papers 1–8 build the prerequisite ladder: thermodynamic framing, geochemical energy and transport, oscillating polymerization, autocatalytic closure, and adaptive boundaries. A protocell requires these pieces to become coupled. Replication without a boundary may preserve molecular information but does not necessarily create a lineage. A boundary without inherited internal chemistry can divide physically without biological heredity.

Paper 9 therefore asks two linked questions:

1. When does copying preserve information rather than dissolve into error?
2. When does copying affect the persistence and division of the container that carries it?

2 Template Growth and Replication Gain

For a schematic template T and monomer pool M_i ,

$$T + M_i \rightarrow TM_i, \quad R_{\text{rep}} = k_{\text{rep}} c_T c_M. \quad (1)$$

Transport-limited monomer supply can be represented phenomenologically as

$$\mathbf{J}_M = -\frac{1}{C} \nabla c_M. \quad (2)$$

A well-mixed caricature gives

$$\frac{dc_T}{dt} = k_{\text{rep}}c_Tc_M - \frac{1}{CL^2}c_T = c_T \left(k_{\text{rep}}c_M - \frac{1}{CL^2} \right). \quad (3)$$

Define the replication gain

$$\Gamma_{\text{rep}} = k_{\text{rep}}c_MCL^2. \quad (4)$$

The condition $\Gamma_{\text{rep}} > 1$ is only a linearized growth condition. It is not sufficient for life.

3 Error Threshold

If per-step fidelity is q across n additions, the simplest master-sequence survival approximation is

$$Q = q^n, \quad \epsilon = 1 - Q. \quad (5)$$

Information persists only if replication gain outruns error catastrophe. A schematic compatibility condition is

$$\Gamma_{\text{rep}}Q > 1, \quad (6)$$

with the usual warning that real prebiotic copying is not a clean independent-site process [4, 2].

The CDFD hypothesis is that fidelity, monomer activation, retention, and damage are coupled to energy throughput. A useful bookkeeping score is

$$\mathcal{H}_{\text{copy}} = \Gamma_{\text{rep}}QR_{\text{mono}}R_{\text{ret}}R_{\text{damage}}^{-1}, \quad (7)$$

where R_{mono} is activated-monomer availability, R_{ret} is retention of templates and products, and R_{damage} collects hydrolysis, photodamage, side reactions, and strand loss. This is a checklist, not a calibrated law.

4 Quasispecies Logic Versus Protocell Logic

The quasispecies threshold and the protocell threshold are related but not identical. The first asks whether a sequence distribution preserves information. The second asks whether a bounded domain inherits a coupled chemistry that affects its growth or persistence.

The protocell condition can be stated as a covariance requirement:

$$\text{Cov}(\text{copied internal state}, \text{container persistence}) > 0. \quad (8)$$

Without this covariance, selection may act on free templates or on empty containers, but not yet on a proto-cell lineage.

5 Integrated Protocell Scaffold

An integrated toy protocell has state variables such as

$$U = \{c_i, \Phi_e, \Phi_H, C_i, \dots\}, \quad (9)$$

evolving through transport, reaction, and constraint laws [6]. A minimal transport skeleton is

$$\frac{\partial c_i}{\partial t} = \nabla \cdot \left(\frac{1}{C_i} \nabla c_i \right) + R_i(c, \Phi_e, \Phi_H), \quad (10)$$

with potentials satisfying analogous updates:

$$\frac{\partial \Phi_e}{\partial t} = \nabla \cdot \left(\frac{1}{C_e} \nabla \Phi_e \right) + S_e, \quad (11)$$

$$\frac{\partial \Phi_H}{\partial t} = \nabla \cdot \left(\frac{1}{C_H} \nabla \Phi_H \right) + S_H. \quad (12)$$

Flux-coupled reactions can be written as

$$\mathbf{J}_e = -\frac{1}{C_e} \nabla \Phi_e, \quad \mathbf{J}_H = -\frac{1}{C_H} \nabla \Phi_H, \quad (13)$$

$$R_i = f_i(c) g(|\mathbf{J}_e \cdot \mathbf{J}_H|), \quad (14)$$

for nonnegative g .

6 Numerical Audit

The diagnostic for this paper combines a reduced error-threshold sweep with a spatial protocell update. It should be read as a numerical stress test, not as validation. The reduced diagnostic reported:

- $\Phi = 0.2$: final master-sequence fitness 0.367,
- $\Phi = 0.5$: final master-sequence fitness 0.083,
- $\Phi = 1.0$: final master-sequence fitness 0.014,
- $\Phi = 5.0$: final master-sequence fitness 1.000,
- $\Phi = 10.0$: final master-sequence fitness 1.000.

The non-monotonic low-to-intermediate response rules out a single universal critical flux in this reduced diagnostic. The correct claim is windowed: copying benefits from energy throughput only when activation, retention, and damage remain bounded.

The integrated protocell diagnostic is more important for what it exposes. It produced finite early snapshots and then non-finite behavior under unbounded explicit updates. That failure blocks any claim of a stable simulated protocell.

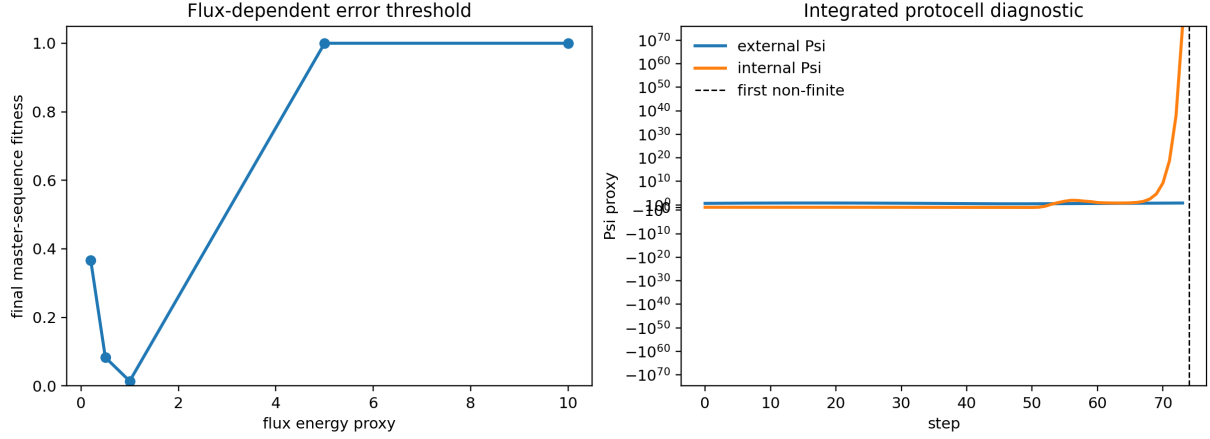


Figure 1: Merged error-threshold/protocell diagnostic. The left panel tests flux-dependent master-sequence persistence. The right panel records numerical instability in the integrated protocell update. The figure is a stress test, not a successful protocell simulation.

7 Hardening Roadmap

The next implementation should replace unbounded autocatalytic growth with saturated kinetics such as

$$R_{\text{auto}} = \frac{k\Phi^2}{K + \Phi^2} - \ell\Phi, \quad (15)$$

then verify timestep refinement, boundedness, and convergence. Operator splitting or implicit updates may be needed for coupled transport and reaction.

8 Experimental Targets

The model protocell literature gives concrete targets: template-directed genetic-polymer synthesis in fatty-acid vesicles [3], nonenzymatic RNA synthesis inside model protocells [1], multi-step nonenzymatic primer-extension chemistry [7], and recent water-compatible thioester-mediated RNA aminoacylation/peptidyl-RNA chemistry [5]. CDFD should therefore be tested by a coupled score,

$$\mathcal{R}_{\text{proto}} = \Gamma_{\text{rep}} Q R_{\text{retention}} R_{\text{exchange}} R_{\text{division}}, \quad (16)$$

not by replication alone.

9 Conclusion

Paper 6 is the integration gate. It establishes that bounded copying, retained internal chemistry, coupled energy flow, and heritable containers must be tested together. The current numerical status is deliberately conservative: the reduced copying toy is informative, while the full protocell implementation remains unstable. Paper 10 can only discuss parasitic exploitation after this replication/container machinery exists.

10 Reproducibility Note

The companion script `supplementary_ool_09.py` writes `outputs/paper_09/error_threshold_and_protocell` plus CSV/JSON summaries. No notebook is required.

References

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